

C started with $R_1 = c$ eventually halts

if and only if

H' started with $R_1 = c$ halts with $R_0 = 0$

if and only if

H started with $R_1 = c, R_2 = \lceil c \rceil$ halts with $R_0 = 0$

if and only if

$prog(c)$ started with $R_1 = c$ does not halt

if and only if

C started with $R_1 = c$ does not halt

Contradiction!